

Create database and create table

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with a search filter. The main editor window contains the following SQL script:

```
1 • CREATE DATABASE bookflow4_db;
2 • USE bookflow4_db;
3 • CREATE TABLE Books
4 • (
5 •     book_id INT AUTO_INCREMENT PRIMARY KEY,
6 •
7 •     title VARCHAR(200) NOT NULL,
8 •
9 •     author VARCHAR(150) NOT NULL,
10 •
11 •     genre VARCHAR(100),
12 •
13 •     description TEXT,
14 •
15 •     published_year INT
16 • );
17
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom panel shows the 'Output' window with the following table:

#	Time	Action	Message	Duration / Fetch
1	12:42:39	CREATE DATABASE bookflow4_db	1 row(s) affected	0.000 sec
2	12:42:40	USE bookflow4_db	0 row(s) affected	0.000 sec
3	12:42:40	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(150) NOT NULL, genre VARCHAR(100), description TEXT, published_year INT)	0 row(s) affected	0.031 sec

The screenshot shows the MySQL Workbench interface after executing the 'DESCRIBE Books' command. The main editor window contains the following SQL script:

```
10 •
11 •     genre VARCHAR(100),
12 •
13 •     description TEXT,
14 •
15 •     published_year INT
16 • );
17
18 • DESCRIBE Books;
19
```

The right sidebar shows the same message as the previous screenshot.

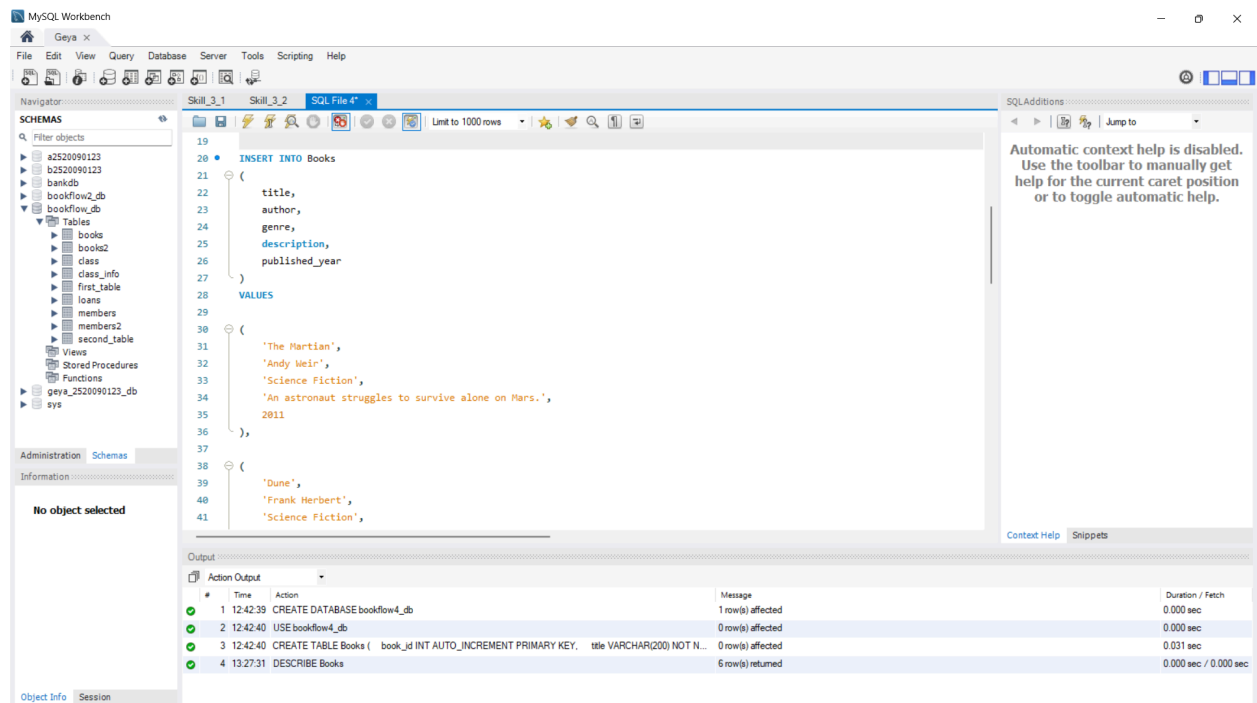
The bottom panel shows the 'Result Grid' window with the following table:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PK		auto_increment
title	varchar(200)	NO			
author	varchar(150)	NO			
genre	varchar(100)	YES			
description	text	YES			
published_year	int	YES			

The bottom panel also shows the 'Output' window with the following table:

#	Time	Action	Message	Duration / Fetch
1	12:42:39	CREATE DATABASE bookflow4_db	1 row(s) affected	0.000 sec
2	12:42:40	USE bookflow4_db	0 row(s) affected	0.000 sec
3	12:42:40	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NULL, author VARCHAR(150) NOT NULL, genre VARCHAR(100), description TEXT, published_year INT)	0 row(s) affected	0.031 sec
4	13:27:31	DESCRIBE Books	6 row(s) returned	0.000 sec / 0.000 sec

Insert values into table



MySQL Workbench interface showing the execution of an SQL query. The query is an INSERT INTO statement for the 'Books' table. The output pane displays the execution results, including the creation of the database and table, and the successful insertion of three rows.

```
19
20 INSERT INTO Books
21 (
22     title,
23     author,
24     genre,
25     description,
26     published_year
27 )
28 VALUES
29 (
30     'The Martian',
31     'Andy Weir',
32     'Science Fiction',
33     'An astronaut struggles to survive alone on Mars.',
34     2011
35 ),
36 (
37     'Dune',
38     'Frank Herbert',
39     'Science Fiction',
40     'A desert planet war',
41     1965
42 ),
43 (
44     'Pride and Prejudice',
45     'Jane Austen',
46     'Romance',
47     'A story of love and class in 19th-century England.',
48     1813
49 ),
50 (
51     'The Great Gatsby',
52     'F. Scott Fitzgerald',
53     'Classic',
54     'A story of love and class in 1920s New York.',
55     1925
56 ),
57 (
58     'Foundation',
59     'Isaac Asimov',
60     'Science Fiction',
61     'A story of the rise and fall of a galactic empire.',
62     1951
63 ),
64 (
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66     'J.R.R. Tolkien',
67     'Fantasy',
68     'A story of a hobbit's journey to a hidden world.',
69     1937
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73     'Orson Scott Card',
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75     'A story of a young boy's journey to become a military leader.',
76     1985
77 ),
78 (
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81     'Fantasy',
82     'A story of a young wizard's journey to Hogwarts.',
83     1997
84 ),
85 (
86     'Twenty Thousand Leagues Under the Seas',
87     'Jules Verne',
88     'Adventure',
89     'A story of a submarine journey to the bottom of the ocean.',
90     1870
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852     'A story of a young wizard's journey to Hogwarts.',
853     1997
854 ),
855 (
856     'Twenty Thousand Leagues Under the Seas',
857     'Jules Verne',
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860     1870
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874     1965
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880     'A story of a man's journey to Mars.',
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884     'Pride and Prejudice',
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891     'The Great Gatsby',
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898     'Foundation',
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901     'A story of the rise and fall of a galactic empire.',
902     1951
903 ),
904 (
905     'The Hobbit',
906     'J.R.R. Tolkien',
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908     'A story of a hobbit's journey to a hidden world.',
909     1937
910 ),
911 (
912     'Ender's Game',
913     'Orson Scott Card',
914     'Science Fiction',
915     'A story of a young boy's journey to become a military leader.',
916     1985
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1220     'Dune',
1221     'Frank Herbert',
1222     'Science Fiction',
1223     'A story of a man's journey to a desert planet.',
1224     1965
1225 ),
1226 (
1
```

Adding embedding column

The screenshot shows the MySQL Workbench interface with the 'SQL File 4' editor open. The editor contains the following SQL script:

```
114 genre,  
115 published_year  
116 FROM Books  
117 ORDER BY book_id;  
118  
119 • ALTER TABLE Books  
120 ADD COLUMN embedding JSON;  
121  
122 • DESCRIBE Books;  
123
```

The 'Result Grid' shows the schema of the 'Books' table:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PK		auto_increment
title	varchar(200)	NO			
author	varchar(150)	NO			
genre	varchar(100)	YES			
description	text	YES			
published_year	int	YES			
embedding	json	YES			

The 'Output' pane shows the execution results of the SQL script:

#	Time	Action	Message	Duration / Fetch
19	13:35:56	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec
20	13:35:56	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Mar...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
21	13:35:56	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0...	20 row(s) returned	0.000 sec / 0.000 sec
22	13:35:56	ALTER TABLE Books ADD COLUMN embedding JSON	Error Code: 1060 Duplicate column name 'embedding'	0.000 sec
23	13:35:56	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec

Inserting mock vectors

The screenshot shows the MySQL Workbench interface with the 'SQL File 4' editor open. The editor contains the following SQL script:

```
161 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)  
162 WHERE book_id = 10;  
163  
164 • SELECT  
165 book_id,  
166 title,  
167 embedding  
168 FROM Books  
169 ORDER BY book_id;  
170
```

The 'Result Grid' shows the schema of the 'Books' table:

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhiker's Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
11	The Martian	
12	Dune	

The 'Output' pane shows the execution results of the SQL script:

#	Time	Action	Message	Duration / Fetch
39	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
40	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
41	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
42	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
43	13:39:05	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

Geya x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

a2520090123
b2520090123
bankdb
bookflow_db
bookflow_db
Tables
books
books2
class
class_info
first_table
loans
members
members2
second_table
Views
Stored Procedures
Functions
geya_2520090123_db
sys

Administration Schemas

Information

No object selected

SQL File 4

```

171 SELECT
172     book_id, title, author, genre,
173     ROUND(
174         SQRT(
175             POW(
176                 CAST(
177                     JSON_EXTRACT(embedding, '$[0]')
178                     AS DECIMAL(10,6)) - 0.90, 2)
179             +
180             POW(
181                 CAST(
182                     JSON_EXTRACT(embedding, '$[1]')
183                     AS DECIMAL(10,6)) - 0.90, 2)
184         ), 2) AS l2_distance
185 FROM Books
186 WHERE embedding IS NOT NULL
187 ORDER BY l2_distance ASC
188 LIMIT 3;

```

Result Grid

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

Output

Action Output

#	Time	Action	Message	Duration / Fetch
40	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
41	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
42	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
43	13:39:05	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
44	13:42:10	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(...	3 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Cosine Similarity

MySQL Workbench

Geya x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

a2520090123
b2520090123
bankdb
bookflow_db
bookflow_db
Tables
books
books2
class
class_info
first_table
loans
members
members2
second_table
Views
Stored Procedures
Functions
geya_2520090123_db
sys

Administration Schemas

Information

No object selected

SQL File 4

```

194 FROM Books
195 WHERE embedding IS NOT NULL
196 ORDER BY l2_distance ASC
197 LIMIT 3;
198
199 SELECT
200     book_id,
201     title,
202     cosine_similarity
203 FROM Books
204 WHERE embedding IS NOT NULL
205 ORDER BY cosine_similarity DESC
206 LIMIT 3;

```

Result Grid

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Output

Action Output

#	Time	Action	Message	Duration / Fetch
41	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
42	13:39:05	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
43	13:39:05	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
44	13:42:10	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(...	3 row(s) returned	0.000 sec / 0.000 sec
45	13:45:24	SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTR...	3 row(s) returned	0.016 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.